

**CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY**  
**Third Semester of M. Sc. (Mathematics) Examination November 2018**  
**MA817 : Mathematical Foundation of Mechanics**

**Date : 01/12/2018, Saturday**

**Time: 10:00 a.m. to 10:45 a.m.**

**Total marks 20**

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**Multiple Choice Questions**

**Instructions:**

1. Tick the correct answer only in this sheet.
2. You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
3. Maximum time to answer Multiple Choice Questions will be 45 minutes.
4. Each question carries one mark.
5. All notations and terminologies are standard.

**Q-I Choose the correct answer from the given options in the following:**

**[20]**

- (1) Degree of freedom for a spherical pendulum is \_\_\_\_\_.  
(a) 0 (b) 1  
(c) 2 (d) 6
- (2) Which of the following gives Euler-Lagrange equation?  
(a)  $\frac{\partial f}{\partial x_i} = \frac{d}{dx_i} \left( \frac{\partial f}{\partial x_i} \right)$  (b)  $\frac{\partial f}{\partial y_i} = \frac{d}{dx_i} \left( \frac{\partial f}{\partial y_i} \right)$   
(c)  $\frac{\partial f}{\partial y_i} = \frac{d}{dx} \left( \frac{\partial f}{\partial \dot{y}_i} \right)$  (d)  $\frac{\partial f}{\partial \dot{x}_i} = \frac{d}{dx_i} \left( \frac{\partial f}{\partial \dot{x}_i} \right)$
- (3) Which of the following statement is not correct?  
(a)  $\frac{\partial H}{\partial t} = \frac{dh}{dt}$  (b)  $H = h$

- (c)  $\frac{dL}{dt} = \frac{dH}{dt}$  (d) None of these
- (4)  $\{q_1, q_2\}$  is \_\_\_\_\_.  
 (a) a Fundamental Lagrange bracket (b) a Fundamental Poisson bracket  
 (c) a zero matrix (d) an undefined term
- (5) Which of the following statement is not true?  
 (a) Geodesics on a sphere are great circles.  
 (b) Cycloid is a Brachistochrone curve.  
 (c) Catenary is the curve of shortest distance.  
 (d) None of these
- (6) Which of the following statement is incorrect?  
 (a) Inverse of a canonical transformation is canonical.  
 (b) Poisson brackets are symmetric in their arguments.  
 (c) Identity matrix is symplectic.  
 (d) None of these
- (7) Which of the following statement is correct?  
 (a)  $\frac{dL}{dt} = 0$ .  
 (b) Lagrangian does not exist if frictional forces are present.  
 (c) Lagrangian may not be unique.  
 (d) None of these
- (8) For a Jacobian Matrix  $M$  of a canonical transformation, which of the following statement is incorrect?  
 (a)  $M$  is identity. (b)  $|M| = \pm 1$ .  
 (c)  $MJM^T = J$ . (d)  $M^{-1}$  is symplectic.
- (9) If all the coordinates of a system are non-cyclic, then Routhian  $R =$  \_\_\_\_\_.  
 (a)  $L$  (b)  $H$   
 (c)  $-L$  (d)  $-H$
- (10) If  $\frac{\partial H}{\partial t} = 0$  then \_\_\_\_\_.  
 (a) Lagrangian is conserved (b) Hamiltonian is conserved  
 (c) all momenta are conserved (d) nothing is conserved
- (11) What are geodesics on the surface of sphere?  
 (a) Straight lines (b) Helix  
 (c) Great circles (d) Large circles
- (12) If Hamiltonian does not depend on a coordinate  $q_j$  explicitly, then \_\_\_\_\_.  
 (a)  $\dot{p}_j = 0$  (b)  $\dot{p}_j \neq 0$   
 (c)  $\frac{\partial H}{\partial \dot{q}_j} = 0$  (d)  $\frac{\partial H}{\partial \dot{q}_j} \neq 0$
- (13) Which of the following statement is incorrect?  
 (a)  $\{u, v\} = -\{v, u\}$ .  
 (b)  $[u, v] = -[v, u]$ .  
 (c) Lagrange bracket vanishes identically.  
 (d) Fundamental Lagrange brackets are canonical invariant.
- (14)  $[\eta, \eta] =$  \_\_\_\_\_.  
 (a)  $J$  (b)  $M$   
 (c) zero matrix (d) not defined
- (15) For a system of  $N$  particles having  $n$  degrees of freedom \_\_\_\_\_.  
 (a)  $N < 3n$  (b)  $n \leq 3N$   
 (c)  $n = 3N$  (d)  $n < N$

- (16) For a system of 5-degrees of freedom, the dimension of phase space is \_\_\_\_\_.  
 (a) 10 (b) 1  
 (c) 5 (d) 25
- (17) Motion of a particle in plane is \_\_\_\_\_ constraint.  
 (a) a non holonomic (b) a rheonomic  
 (c) a holonomic (d) not a
- (18) Jacobian matrix for a canonical transformation is \_\_\_\_\_.  
 (a) Identity (b) equal to  $J$   
 (c) zero matrix (d) symplectic
- (19) A force is called conservative if \_\_\_\_\_ for some scalar  $V$ .  
 (a)  $\vec{F} = -\nabla V$  (b)  $\vec{F} = \alpha V$   
 (c)  $\vec{F} = \nabla \times V$  (d)  $\vec{F} = V$
- (20) For a system of 3 degrees of freedom, the Poisson bracket  
 $[2q_1 + 5q_3, 3q_2] =$  \_\_\_\_\_.  
 (a) 0 (b) 10  
 (c) 6 (d) -6

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**Date:01/12/2018,Saturday Time:10:45 a.m. to 1:00 p.m. Total marks of Section I & II 50**

**Instructions:**

6. Section I and Section II must be written in separate answer books.
7. You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
8. Figures to the right indicate marks.
9. All notations and terminologies are standard.

**SECTION - I**

**Q-II (A) Attempt any TWO of the following: [8]**

- (1) Discuss the law of conservation of total linear momentum for a system of  $N$  number of particles.

- (2) Find the energy function for a system with Lagrangian

$$L = \frac{I_1}{2} (\dot{\theta}^2 + \dot{\phi}^2 \sin^2 \theta) + \frac{I_3}{2} (\dot{\psi} + \dot{\phi} \cos \theta)^2 - mgl \cos \theta,$$

where  $\theta, \phi, \psi$  are generalized coordinates.

- (3) Obtain Routhian equations of motion for a system with Lagrangian

$$L = \frac{m}{2} (\dot{r}^2 + r^2 \dot{\theta}^2) + \frac{k}{r},$$

where  $r, \theta$  are generalized coordinates.

**(B) Attempt any SIX of the following: [12]**

- (1) State and prove the law of conservation of linear momentum for a particle.
- (2) State and prove the principle of virtual work for a system of particles.
- (3) Show that the generalized momentum conjugate to a cyclic coordinate is conserved.
- (4) Define cyclic coordinate and generalized momentum.
- (5) If  $H$  is Hamiltonian of a system, then show that  $\frac{dH}{dt} = \frac{\partial H}{\partial t}$ .
- (6) Define ignorable coordinate and Phase space.
- (7) Compute Routhian for the Hamiltonian  $H = \frac{p^2}{4} + 2q^2$ .
- (8) Define Fundamental Poisson Brackets and find its values.
- (9) If  $u$  is a dynamical quantity which does not depend on  $t$  explicitly, then prove that  $u(t) = (u)_{t=0} + t[u, H]_{t=0} + \frac{t^2}{2!} [[u, H], H]_{t=0} + \dots$

## SECTION – II

**Q-III Answer the following:**

**(A) Attempt any *THREE* of the following:**

**[18]**

- (1) Obtain Lagrange's equation of motion in general form.
- (2) Define canonical variables. Derive Hamilton's equations of motion from Hamilton's modified principle.
- (3) State and prove law of conservation of linear momentum for a system of N-particles in Lagrangian formalism.
- (4) Show that the matrix for infinitesimal canonical transformation is symplectic.
- (5) (i) Discuss D'Alembert's principle for a system of N-particles.  
(ii) Derive Lagrange's equation of motion from Hamilton's principle.

**(B) Attempt any *THREE* of the following:**

**[12]**

- (1) Obtain Lagrange's equation of motion for a particle moving in the space under a force  $\vec{F}$  using Cartesian coordinates.
- (2) State and prove the law of conservation of energy in Lagrangian formalism.
- (3) Discuss matrix form of the Hamilton's equations of motion.
- (4) Derive canonical transformation generated by  $F_2(q, P, t)$  and  $F_3(p, Q, t)$ .
- (5) State and prove Poisson's theorem.

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